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# MINI PROJECT

## TITANIC SURVIVAL PREDICTION USING DECISION TREE

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# TITANIC SURVIVAL PREDICTION USING DECISION TREE

## 1. INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions without being explicitly programmed. Classification is one of the most widely used Machine Learning techniques used to predict categorical outputs.

In this project, a Decision Tree Classifier is used to predict whether a passenger survived the Titanic disaster based on features such as gender, age, and passenger class.

The Titanic dataset is one of the most popular beginner-level datasets in Data Science and Machine Learning because it contains real-world passenger information and survival outcomes.

This project demonstrates the complete Machine Learning workflow including data preprocessing, handling missing values, encoding categorical data, model training, prediction, evaluation, and deployment.

## 2. OBJECTIVE

The main objectives of this project are:

- To analyze Titanic passenger data
- To identify factors affecting passenger survival
- To build a Machine Learning classification model
- To predict survival outcomes accurately
- To understand Decision Tree algorithms practically

## 3. TECHNOLOGIES USED

| Technology | Purpose |
|---|---|
| **Python** | High-level programming language used as the primary tool for data science. |
| **Pandas** | Essential library for data manipulation, cleaning, and analysis using DataFrames. |
| **NumPy** | Supports large, multi-dimensional arrays and matrices, along with high-level mathematical functions. |
| **Scikit-learn** | The go-to library for implementing machine learning models (classification, regression, clustering). |
| **Matplotlib** | A comprehensive library for creating static, animated, and interactive visualizations in Python. |

## 4. DATASET DESCRIPTION

The Titanic dataset contains passenger details from the Titanic ship disaster.

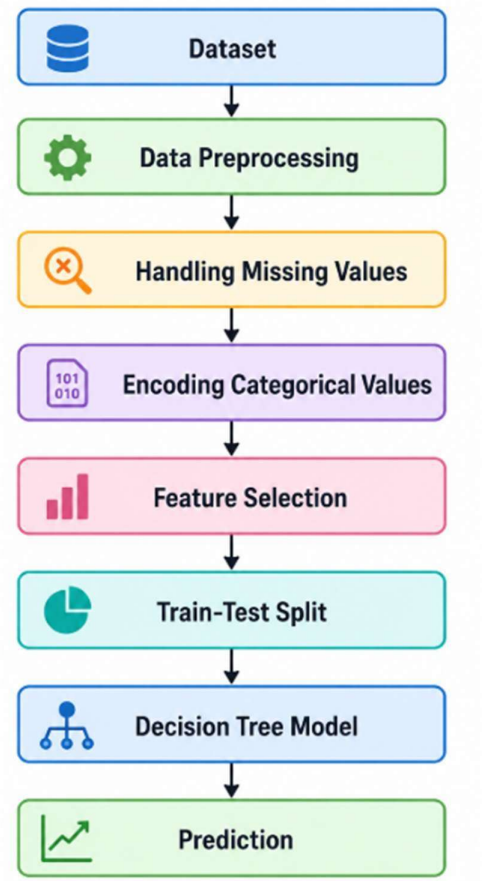
### Features Used

| Feature | Description | Data Type (Typical) |
|---|---|---|
| sex | Gender of the passenger (often a key predictor for survival). | Categorical (String/Boolean) |
| age | Age of the passenger; often contains missing values that require preprocessing. | Numerical (Float/Integer) |
| class | Passenger social-economic status (1st, 2nd, or 3rd class). | Ordinal (Categorical) |

### Target Variable

| Target | Value Mapping | Description |
|---|---|---|
| **survived** | 0, 1 | Indicates the survival status of the passenger. |
| | **0** | The passenger did **not** survive. |
| | **1** | The passenger **survived**. |

## 5. SYSTEM ARCHITECTURE




## 6. DATA PREPROCESSING

Data preprocessing is one of the most important steps in Machine Learning because real-world datasets often contain missing values and categorical data.

The Titanic dataset was loaded using Pandas.

```
import pandas as pd

df = pd.read_csv("Titanic.csv")
```

## 7. HANDLING MISSING VALUES

The age column contained missing values.

To handle missing data, the SimpleImputer class from Scikit-learn was used with the mean strategy.

```
from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy='mean')

df['age'] = imputer.fit_transform(df[['age']])
```

### Why Mean Imputation?

Mean imputation replaces missing values using the average value of the column.

Advantages:

- Prevents data loss
- Maintains dataset size
- Improves model training

Missing values were checked using:

```
df.isnull().sum()
```

## 8. ENCODING CATEGORICAL VALUES

Machine Learning models cannot directly process text data. Therefore categorical columns were converted into numerical values using Label Encoding.

```
from sklearn.preprocessing import LabelEncoder

le1 = LabelEncoder()
le2 = LabelEncoder()

df['sex'] = le1.fit_transform(df['sex'])
df['class'] = le2.fit_transform(df['class'])
```

### Encoded Values

### Sex Encoding

| Original Value | Encoded Value |
|---|---|
| **Female** | **0** |
| **Male** | **1** |

## Passenger Class Encoding

| Original Class | Encoded Value | Description |
|---|---|---|
| **First** | 0 | Premium/Upper Class |
| **Second** | 1 | Middle Class |
| **Third** | 2 | Standard/Lower Class |

## 9. FEATURE SELECTION

The selected input features are:

```
x = df[['sex', 'age', 'class']]
```

Target variable:

```
y = df['survived']
```

## 10. TRAIN-TEST SPLIT

The dataset was divided into training and testing sets.

```
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.2,
    random_state=42
)
```

### Purpose of Train-Test Split

- Training data is used to train the model
- Testing data is used to evaluate performance

## 11. DECISION TREE CLASSIFIER

A Decision Tree is a supervised Machine Learning algorithm used for classification and prediction.

It works like a flowchart where decisions are made based on conditions.

Example:

Is passenger female?

├── Yes → Higher survival chance

└── No → Lower survival chance

The model automatically learns decision rules from the dataset.

## 12. MODEL TRAINING

The Decision Tree model was trained using Scikit-learn.

```
from sklearn.tree import DecisionTreeClassifier

classifier = DecisionTreeClassifier()

classifier.fit(x_train, y_train)
```

## 13. PREDICTION

Predictions were made using:

```
y_pred = classifier.predict(x_test)
```

## 14. MODEL EVALUATION

The model accuracy was calculated using:

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print(accuracy)
```

### Accuracy Achieved

Approximately **81% accuracy**

## 15. KEY INSIGHTS

### 1. Female passengers had higher survival rates

Women were prioritized during rescue operations.

### 2. First-class passengers survived more

Higher-class passengers had better access to lifeboats.

### 3. Younger passengers had better survival chances

Children and younger passengers received rescue priority.

## 16. ADVANTAGES

- Beginner-friendly Machine Learning project
- Demonstrates complete ML workflow
- Real-world dataset analysis

## 17. LIMITATIONS

- Limited number of features
- Basic Decision Tree may overfit
- Accuracy can be improved using advanced models

## 18. FUTURE ENHANCEMENTS

Future improvements can include:

- Random Forest Classifier
- XGBoost
- Hyperparameter tuning
- Streamlit frontend
- Deep Learning models
- Real-time prediction interface

## 19. CONCLUSION

This project successfully demonstrated how Machine Learning can predict passenger survival using the Titanic dataset.

The project provided practical experience in:

- Data preprocessing
- Missing value handling
- Label encoding
- Decision Tree model building
- Prediction and evaluation

The Decision Tree model achieved approximately **81% accuracy**, showing good predictive performance.